Microsoft Azure Implementation - Big Data, GenAI & Agentic AI Architecture

Data sources

Applications

Data ingestion

Batch data
Azure Data Factory
AzCopy · Synapse Pipelines

Real-time streaming
Event Hubs · Event Grid
Stream Analytics · Fabric RTI

Change data capture
Data Factory CDC · Fabric Mirroring
Azure DMS · Debezium on AKS

Reverse ETL
Data Factory · Logic Apps
Azure Functions · Event Grid

Data Ocean

Data lake
ADLS Gen2 + Fabric OneLake

Data pond
Fabric Workspace · ADLS container

Data puddle
ADLS project folder + RBAC

Data mesh
Microsoft Purview · Fabric Domains

Data contracts
Purview Schema Registry · Event Hubs

Workflow manager · Data Factory · Synapse Pipelines · Logic Apps · Durable Functions

Data transformation

Data labeling
Azure ML Data Labeling

Query engines
Synapse · Databricks · Fabric SQL

Data sci. libraries
Azure ML Studio · Fabric Notebooks

Data enrichment
Azure Data Share · Data Factory

AI infrastructure

Knowledge store

Feature store
Azure ML Feature Store

Vector database
AI Search Vector · Cosmos DB
PostgreSQL pgvector

Knowledge graph
Cosmos DB Gremlin · Fabric GraphQL

LLMOps + MLOps

Model registry
Azure ML Model Registry

Prompt registry
AI Foundry Prompt Flow

Fine-tuning · RLHF
Azure OpenAI Fine-tuning · Azure ML

Eval & red-teaming
AI Foundry Evaluations · PyRIT

CI/CD · experiment tracking
Azure DevOps · GitHub Actions · MLflow

AI inference

Online predictor
Azure ML Online Endpoints · AKS

Batch predictor
Azure ML Batch Endpoints · Synapse

Streaming inference
Stream Analytics + Azure ML · Functions

Edge inference
Azure IoT Edge · Azure Stack Edge

AI gateway
APIM AI Gateway · Front Door

GenAI platform

Guardrails & orchestration
Azure AI Content Safety · Prompt Flow

Model & prompt hub
AI Foundry Model Catalog

Custom knowledge model
Azure OpenAI Fine-tuning · Azure ML

Foundation models
Azure OpenAI Service · AI Foundry
GPT · Phi · Llama · Mistral · DeepSeek

RAG pipeline
Azure AI Search (vector + hybrid)
On Your Data · Prompt Flow

Multimodal
GPT-4o · AI Vision · Speech
Document Intelligence · Translator

Agentic AI layer

Agent orchestrator
AI Foundry Agent Service
Semantic Kernel · AutoGen
Copilot Studio

Tool registry / MCP
AI Foundry Tools · MCP server
Azure Functions · API Mgmt

Agent memory
Cosmos DB · Azure Cache for Redis
AI Search · Table Storage

HITL interface
Copilot Studio approvals
Power Automate · Logic Apps

Agent monitoring
Application Insights · Azure Monitor
AI Foundry Tracing

Analytics & BI

Data viz & storyboard
Power BI · Fabric Reports

Data search
Azure AI Search · Microsoft Search

Data accumulation
Synapse · Fabric Warehouse · Cosmos DB

Content analysis
AI Language · Document Intelligence
Translator · Speech

Decision management
Logic Apps · Durable Functions
Personalizer · Azure ML

Agentic analytics
Copilot in Power BI · Fabric Copilot
M365 Copilot

AI trust, governance & security

Data governance
Microsoft Purview · Fabric Data Catalog
Microsoft Defender for Cloud · Key Vault
Entra ID · Activity Log · Azure Policy · Sentinel

AI governance
Azure AI Content Safety · Responsible AI dashboard
Purview AI Hub · Defender for AI · Azure Monitor
Model cards · Azure Policy for AI

Assessment framework

AI maturity assessment
Well-Architected Framework · Cloud Adoption Framework

Economic benefit assessment
Cost Management · Pricing Calculator · Advisor

Responsible AI KPIs
Responsible AI dashboard · Azure ML metrics
Azure Monitor · AI Foundry Evaluations

Data layers | AI/ML infrastructure | GenAI & analytics | Agentic AI | Governance & trust | Assessment